

NoiseGuard AI — Environmental Noise Analysis Report

Generated: 2026-07-18 18:21 UTC (example) | Analysis ID: example-0001

1. Summary

Source file	courtyard_construction_evening.wav
Duration	6.40 s
Top classification	Jackhammer (74.2% confidence)
Relative loudness	Loud (-14.3 dBFS)

2. AI Classification (YAMNet)

Class	Confidence
Jackhammer	74.2%
Power tool	11.8%
Construction	7.1%
Drill	4.0%
Engine	2.9%

3. Acoustic Features (DSP)

RMS (mean / std)	0.1923 / 0.0511
Zero crossing rate	0.0872
Spectral centroid	2143.7 Hz
Spectral bandwidth	1801.2 Hz
Spectral roll-off	4620.5 Hz
Mel-spectrogram mean	-31.4 dB
Estimated relative level	-14.3 dBFS

4. Environmental Interpretation

Detected class: 'Jackhammer'. The dominant sound matches construction activity (drilling, hammering, power tools). This pattern is common in disputes over construction-site noise regulations. The estimated relative intensity is high.

5. Recommendations

- Verify the construction site's permitted working hours under local regulations.
- Record multiple samples across different days to demonstrate a consistent pattern.
- File a complaint with the municipal construction supervision authority, attaching this report.

- For a formally admissible measurement, request an inspection with a certified sound level meter from the relevant sanitary or municipal authority.

6. Important Limitation

This report is generated from an uncalibrated smartphone/microphone recording. The loudness value is a RELATIVE estimate derived from digital signal amplitude (dBFS), NOT a certified decibel (dB SPL) measurement. It must not be treated as legally certified acoustic evidence. For official measurements, contact an accredited acoustic measurement authority.